

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bengaluru-560078

(An Autonomous Institution affiliated to VTU, Belagavi)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Introduction to AI and Applications

CIA-3-Question Bank

Module-4: Introduction to Prompt Engineering

- Define Prompt Engineering and explain its importance in Large Language Model with suitable examples.
- Trace the evolution of Prompt Engineering from rule-based systems to large language models like ChatGPT. Highlight key milestones.
- Explain the different types of prompts used in Prompt Engineering. Illustrate any three types with examples.
- How does Prompt Engineering work in AI systems? Describe the interaction between user prompts and large language models.
- Discuss the role of Prompt Engineering in improving human–AI communication. Explain how prompt design affects response quality.
- Explain the functional components of an effective prompt. What characteristics make a prompt clear, specific, and goal-oriented?
- Analyze the advantages of Prompt Engineering in areas such as Coding, education, content creation, and problem solving.
- Compare poorly designed prompts with well-engineered prompts. Explain how prompt structure impacts AI outputs.
- How does LLM work ? Summarize how prompt engineering improves precision and creativity in LLM outputs.
- Explain the working of Large language model. Describe the significance of the Transformer architecture in the growth of prompt engineering.
- Apply the principles of prompt engineering to improve the following prompt: “Write about AI.”
- Create a detailed prompt with specificity that helps an LLM generate Python code for finding the largest number in a list.
- Develop two different prompts for the same task and compare how specificity changes the output quality.

Module–5: Prompt Engineering Techniques for ChatGPT

- Explain Prompt Engineering techniques used for ChatGPT. Why are specialized techniques required for conversational AI?
- Describe the Instruction Prompt Technique with suitable examples. Explain how it improves task accuracy.
- Explain Zero-shot, One-shot, and Few-shot prompting. Compare them with respect to learning efficiency and response quality.
- Discuss Few-shot prompting in detail. How does it help ChatGPT understand patterns and context?
- Explain the Self-Consistency Prompting technique. How does it improve reasoning and decision-making in AI responses?
- Explain how prompts can be used to unlock imagination and innovation. Design suitable prompts for creative problem-solving.
- Discuss how prompts help ignite the writing process. Create prompts for generating academic, technical, or creative content.
- A product design team is struggling to generate innovative ideas for an eco-friendly product. Design suitable AI prompts to stimulate creative thinking and explain their effectiveness.
- A student is unable to begin writing a technical article due to a lack of structure. Formulate effective prompts to help initiate and organize the writing process.
- An AI-based learning platform aims to support both creative idea generation and effective writing. Design one prompt for creative thinking and one for effective writing. Explain how each achieves its purpose.